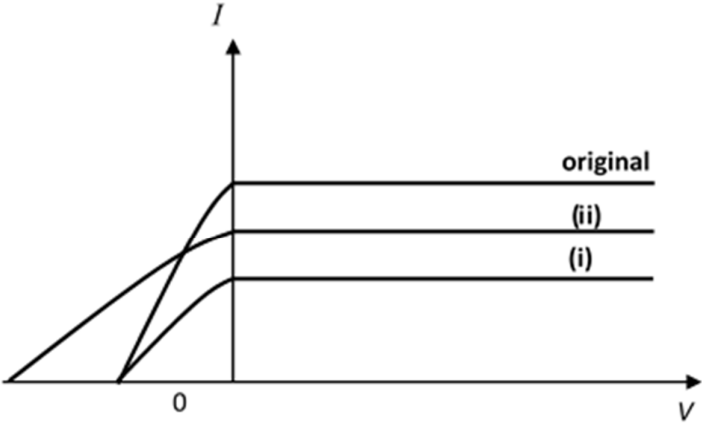


6 (a)	Max KE possessed by electron emitted from surface layer	B1
	Electrons below the surface loses energy along the way as they move towards surface, so less KE than KE_{max}	B1

(b)(i)	 ½ intensity → current is reduced by half	B1
	same stopping potential	B1
(b)(ii)	wavelength is decreased but intensity is constant → number of photons is reduced so current decreases. Can decrease below (i) current.	B1
	but higher stopping potential	B1
(b)(iii)1	$n = \frac{I}{e} = \frac{4.8 \times 10^{-10}}{1.6 \times 10^{-19}} = 3.0 \times 10^9 \text{ s}^{-1}$	A1
(b)(iii)2	The intensity, i $i = N \left(\frac{hc}{\lambda} \right) \left(\frac{1}{A} \right)$ where N = number of photons incident per second = 2500n	C1
	$i = \frac{Nhc}{A\lambda} = \frac{2500(3.0 \times 10^9)(6.63 \times 10^{-34})(3.0 \times 10^8)}{(24 \times 10^{-6})(410 \times 10^{-9})}$	C1
	= 0.152 W m ⁻²	A1

3 (a)	Let the mass of the satellite be m	
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